4. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

CODING:

```
import math

import random


random.seed(1)


def sigmoid(x):
   return 1 / (1 + math.exp(-x))


def dsigmoid(y):
   return y * (1 - y)


X = [[0, 0], [0, 1], [1, 0], [1, 1]]

Y = [0, 1, 1, 0]


w1 = [[random.uniform(-1, 1) for _ in range(2)]
    for _ in range(4)]

w2 = [random.uniform(-1, 1) for _ in range(4)]

b1 = [random.uniform(-1, 1) for _ in range(4)]

b2 = random.uniform(-1, 1)

lr = 0.5


for epoch in range(20000):
   for x, target in zip(X, Y):
      hidden = [
```

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```
    sigmoid(sum(x[j] * w1[i][j] for j in range(2)) + b1[i])
    for i in range(4)
]
output = sigmoid(sum(hidden[i] * w2[i] for i in range(4)) + b2)

error = target - output
delta = error * dsigmoid(output)

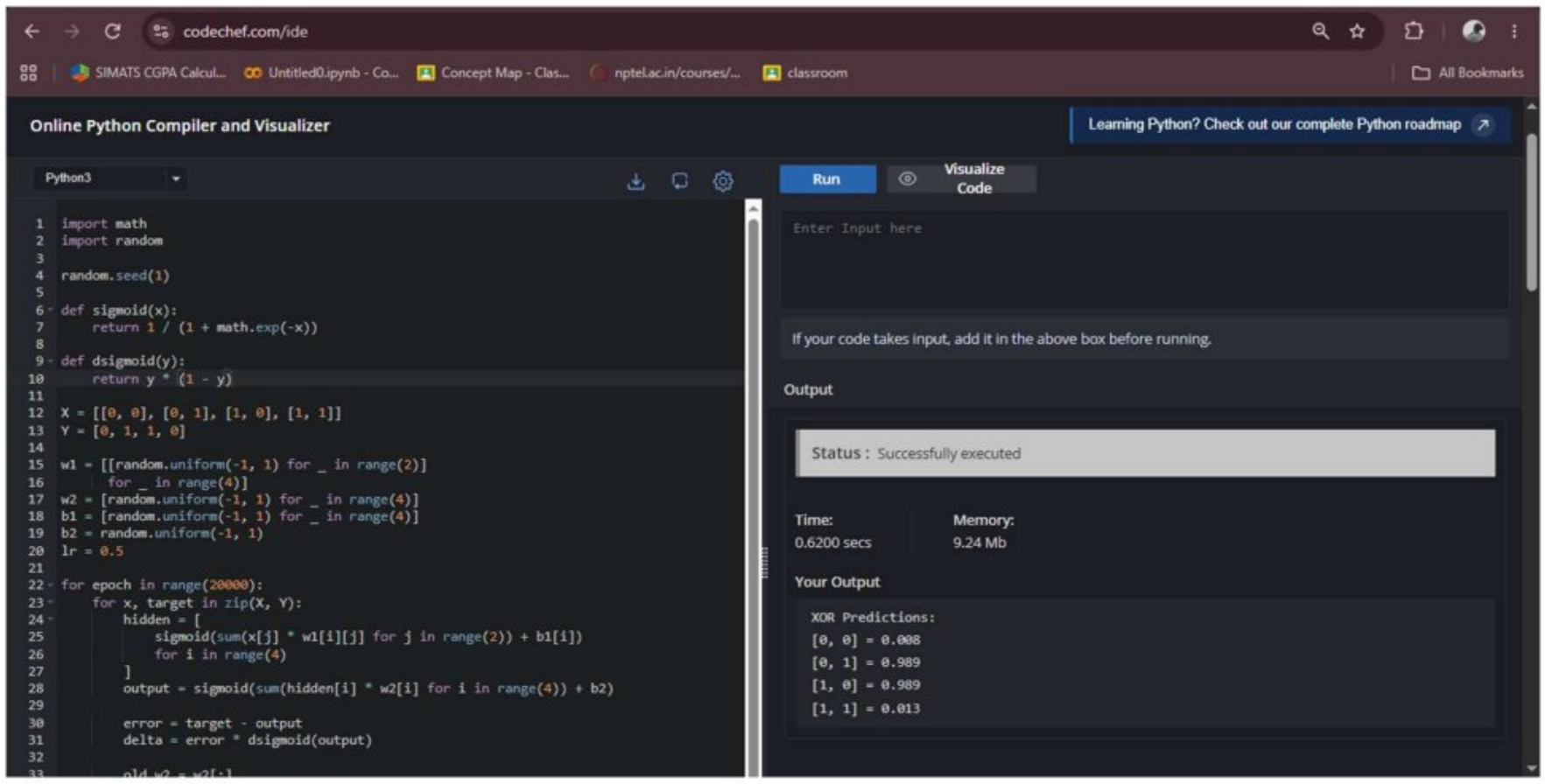
old_w2 = w2[:]
    for i in range(4):
        w2[i] += lr * delta * hidden[i]
    b2 += lr * delta

    for i in range(4):
        d = delta * old_w2[i] * dsigmoid(hidden[i])
        for j in range(2):
            w1[i][j] += lr * d * x[j]
        b1[i] += lr * d

print("XOR Predictions:")
for x in X:
    hidden = [
        sigmoid(sum(x[j] * w1[i][j] for j in range(2)) + b1[i])
        for i in range(4)
    ]
    output = sigmoid(sum(hidden[i] * w2[i] for i in range(4)) + b2)
    print(x, "=", round(output, 3))
```

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OUPUT:



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